

Syed Suhail Raza

Data Analyst

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GITHUB | LINKEDIN | PORTFOLIO

EDUCATION

DR R M L AVADH UNIVERSITY

M.Sc (MATHS & STATS)

Percentage: 71.04%

2013-01-01 - 2015-12-31

Ayodhya, India

EXPERIENCE

CRITERION TECH PRIVATE LIMITED | JUNIOR DATA SCIENCE ENGINEER

Sarfarazganj, Lucknow, India |

2024-07-01 – Present

Built scalable algorithms with ML (YOLOv5, OpenCV). Worked with Python and SPSS to build and deploy data-driven solutions. Proficient in PowerBI, Advanced Excel, Google Sheets. Experience in developing pipelines for signal processing and analysis (BCG, PPG), foot pressure and gait analysis, and radar-based data analysis.

SKILLS

PROGRAMMING LANGUAGES

Python, JavaScript

LIBRARIES/Frameworks

YOLOv5, PyTorch, Scikit-learn, YOLOv8

TOOLS / PLATFORMS

VS Code, Jupyter Notebook, Google Colab, PowerBI, Advanced Excel, Google Sheets

DATABASES

MySQL

PROJECTS / OPEN-SOURCE

BCG BED-BASED HEALTH MONITORING

Python, Pandas, NumPy, SciPy, Matplotlib, OpenCV, Signal

Filtering, Peak Detection, Plotly, CSV, PDF

Developed a Python pipeline to extract and analyze Heart Rate (HR) and Respiratory Rate (RR) from BCG and PPG (RED/IR) signals. Compared BCG-derived vitals with RED-based PPG signals for accuracy validation. Automated 5-minute interval CSV and PDF reporting.

FOOT PRESSURE AND GAIT ANALYSIS USING SENSOR DATA

Python, Pandas, NumPy, Matplotlib,

Plotly, CSV

Processed and visualized foot pressure sensor data to analyze pressure distribution across different foot regions (forefoot, midfoot, heel, lateral). Generated interactive graphs and performed statistical analysis to identify high/low pressure zones.

PPG DATA ANALYSIS USING LINEAR REGRESSION

Python, Pandas, Scikit-learn, Matplotlib, Linear

Regression

Analyzed PPG data using Python and Linear Regression to classify by age and gender, and estimated patient glucose levels. Results visualized with plots.

AGE AND GENDER PREDICTION USING ML

YOLOv8, Python, PyTorch, Google Colab

Used YOLOv8 to predict age and gender from USB camera input and saved results.

RADAR BASED DATA ANALYSIS

Python, Pandas, NumPy, Matplotlib, OpenCV, CSV, HIS

Analyzed heart rate, respiratory rate, sleep, and movement using radar device data. Processed hexadecimal data from multiple CSV files, converted to per-minute values, and validated against HIS data and video recordings.

GCS MEASUREMENT FOOT SOLE DETECTION Developed an edge-optimized Foot Sole Detection system using YOLOv8-Nano, ONNX, and OpenCV on Raspberry Pi 4

A real-time edge-AI system built on Raspberry Pi 4 using YOLOv8 Nano and ONNX to accurately detect, track, and calculate the spatial positioning of a patient's foot sole.

SKIN TURGOR BODY WATER RETENTION SYSTEM ultralytics opencv-python

This is an AI-powered clinical tool that utilizes YOLOv8 computer vision to automate the 'Skin Turgor (Pinch Test)' analysis. It determines a patient's hydration level by precisely calculating the exact time (in milliseconds) it takes for their skin to return from a 'Tented' state back to a normal 'Flat' state.

PHARMACY STORE MANAGEMENT SYSTEM opencv-python / opencv-contrib-python (cv2), numpy, ultralytics (YOLOv8), paddlepaddle (paddle), paddleocr, torch, thefuzz

This project is a locally hosted, dual-camera smart kiosk system that utilizes YOLO object detection and PaddleOCR to instantly scan, identify, and verify medicine strips and their batch expiry details. It is designed to ensure pharmacy dispensing accuracy by matching physical medicines against a local inventory database in real-time without requiring internet access

CERTIFICATIONS

- AI and Cybersecurity Awareness - **TCS ION**
- Java and Python Programming Mastery - **UDEMY**